MARITIME COMMAND QUALIFICATION STANDARD AND PLAN

Submarine Towed Array Handling Course

NQUAL - AIDI

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Chapters 3, 4: 21 Jun 08

RECORD OF AMENDMENTS

The "Authority" reference indicates approval of the corresponding chapters of this QSP document. Authority for Chapters 1 and 2 of this QSP resides with Chief of Maritime Staff (CMS) DMTE. The Curriculum Control Authority (CCA) for Chapters 3 and 4 is Canadian Forces Naval Operations School. The approval date, the date that the document is checked back in, should also appear in the bottom right corner of every page (the change being made in the footer). The Revision column is intended to provide a short indication of where the change(s) was/were made. Details on the change(s) must be amplified in Annex J "Course History".

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FOREWORD

1. This training document is developed under the authority of the Chief Maritime Staff and in accordance with the following references:

   a. A-P9-050 Series, Canadian Forces Manual of Individual Training and Education;

   b. A-PD-055/PP-001 Occupation Specification (Officer) dated [insert date (if applicable)] ;

   c. A-PD-055/PP-002 Occupation Specification (NCM) dated [insert date (if applicable)] ; and

   d. A-PD-055/PP-001 Occupation Specialty Specification (Officer and NCM) dated [insert date (if applicable)].

2. This publication is effective on receipt. Performance Objectives (POs) are explained fully in Chapter 2. The Lesson Specifications (LSs)/Enabling Objectives (EOs), Enabling Checks (ECs) and Performance Checks (PCs) which specify the standards that the trainee must achieve to become qualified are listed in Chapter 4.

3. Suggestions for amendments may be initiated by anyone. Amendments must be forwarded through the normal chain of command. Specifically, training establishments are to summarize all requested amendments in writing to the applicable approving authority(ies). QSP revision procedures are outlined in the NTI. Final approvals for amendments are as follows:

   a. Chapters 1 and 2 – Chief Maritime Staff, [either DMTE or DMPOR] . Suggested amendments must first have the approval of the Curriculum Control Authority Training Establishment; and

   b. Chapters 3 and 4 – Commandant of [CCA]. Training Establishments that conduct training and are not the CCA must forward suggested amendments to [CCA] for approval.

4. [Old training Standard (TS) Chapter 1]

5. The designated CCA(s) for this training is indicated below:

CCA	PHASE	QUALIFICATION

CHAPTER 1: TRAINING MANAGEMENT DETAILS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY CHIEF MARITIME STAFF/DMTE

AIM

1. The aim of this qualification standard and plan (QSP) is to train naval personnel to perform their duties as members of the Submarine Towed Array Handling Team (STAHT)

2. This QSP is designed to assist instructors in preparing and organizing themselves for the STAHT and should be read prior to commencing any preparation or scheduling. Any suggestions for the improvement of this document should be brought to the attention of the standards division of the Curriculum Control Authority.

COURSE DURATION CALCULATION

3. Course duration is calculated in accordance with MARCORD 9-30.

Actual Training Days (ATT)	14 Days
Training Administrative Time (TAT) (if required)	5 Periods
Base Administrative Time	1 Day
Training Travel Time (if required)	7 Periods
Course Duration (total calendar days)	17 days

RESOURCES

4. Identify key resources required to conduct the training. Additional training establishment resources can be found at Annex H.

REQUIRED COURSE STAFFING				
TITLE	**RANK**	**QUALIFICATION**	**QTY**	**REMARKS**
Course Officer	LT(N)	AIDI	1	
Instructor	MS/PO2/PO1	NET/NWT, AIDI	1	With TASR Experience
Instructor	S1/MS/PO2	ER/ET, AIDI	1	With TA Experience

CRITICAL TRAINING RESOURCES	NO. OF DAYS
Containerized Reeler System, Tulsa Reeler System, Stub Cable, Tow Cable	15
600 Series Barge	15
CRITICAL TRAINING RESOURCES	**NO. OF DAYS**
, Glen Tug, Ville Tug, MCDV	2

COURSE CAPACITY

5. Planning for the conduct of training is based on the following guidelines:

   a. Maximum – 8

   b. Optimum – 6

   c. Minimum – 4

COURSE PREREQUISITES

6. The candidates must:

   a. Be designated as a member of the STAHT East or West or as an Alternate;

   b. NCMs must be Journeyman (QL5) Qualified.

   c. Officers must be Submarine Officer Qualified (AILV).

RELATED DOCUMENTS

7. The following documents are related to this QSP:

   a. A-PD-055-003/PQ-001 (OSS AIDI): Submarine Towed Array Handling.

EVALUATION

8. Program evaluation shall be conducted in accordance with the guidance provided in A-P9-050-000 CFITES Vol. 11 Evaluation and Validation Techniques.

9. Trainee evaluation shall be conducted in accordance with NTI 03/2007(TBP).

VALIDATION

10. The training will be validated every five years in accordance with Naval Training Instructions, based on the task list in Annex B.

QUALIFICATION

11. Successful completion of this course will result in the award of the Submarine Towed Array Handling qualification (Competency Code AIDI).

PRIOR LEARNING ASSESSMENT

12. Prior Learning Assessment shall not be conducted for this course.

SPECIAL INSTRUCTIONS

13. The Training establishment shall liaise with outside agencies for coordination of assets and guest lecturers as required to conduct the course.

14. Incremental Staff: During the sea portion of the training up to 7 line handlers will be required, 4 on the barge and 3 on the Ville Tug.

CHAPTER 2: PERFORMANCE OBJECTIVES

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY CHIEF MARITIME STAFF/DMTE

PO 001 PERFORM ALONGSIDE PRE/POST DEPLOYMENT ACTIVITIES FOR SUBMARINE TOWED ARRAY OPERATIONS

1. Performance: Perform Alongside Pre/Post Deployment Activities for Submarine Towed Array Operations

2. Conditions:

   a. Given:

      (1) Tulsa Reeler;

      (2) Towed Array Reels;

      (3) Towed Array Handling Equipment;

      (4) Towed Array Assembly consisting of:

         (a) Stub;

         (b) Tow Cable;

         (c) Array; and,

         (d) Drogue Rope;

      (5) Crane operator and Riggers;

      (6) Towed Array Vessel;

      (7) Submarine Towed Array Handling Team;

      (8) Diver Support Platform (if available);

      (9) Operational Submarine with functioning 2046 Sonar (if available);

      (10) Paul Bunyan;

      (11) Flat bed trailer;

      (12) Forklift and Operator;

      (13) Associated tools and minor equipment;

      (14) Submarine and Technical Assistance;

      (15) References; and,

      (16) Supervision.

   b. Denied:

      (1) N/A.

c. Environmental:

(1) All weather conditions alongside; and,

(2) Winds speeds IAW local regulations for crane operations.

3. Standard. Perform Alongside Pre/Post Deployment Activities for Submarine Operations to include:

a. Identifying Towed Array equipment;

b. Operating TA Handling Equipment; and,

c. Performing Stub connection activities.

4. References.

A1, A2, A4, A5, A6, A7, A8, B1, C1, C2, D1.

5. Specification Task Numbers.

T0001, T0002, T0006, T0007, T0008, T0009, T0021, and T0022

6. Supporting Skills.

Nil

7. Supporting Knowledge. K0001 – K0015, and K0036

8. Training Limitations.

a. Availability of an operational submarine or suitable replacement vessel;

b. Availability of QHM Assets/Crane support; and,

c. Weather

PO 002 CONDUCT SUBMARINE TOWED ARRAY OPERATIONS AT SEA

1. Performance: Conduct Submarine Towed Array Operations At Sea.

2. Conditions:

   a. Given:

      (1) Tulsa Reeler;

      (2) Towed Array reels;

      (3) Towed Array Handling Equipment;

      (4) Towed Array System consisting of:

         (a) Stub cable;

         (b) Tow Cable;

         (c) Array; and,

         (d) Drogue Rope;

      (5) Towed Array Vessel;

      (6) Submarine Towed Array Handling team;

      (7) Operational Submarine with functioning 2046 sonar or suitable vessel if Submarine not available;

      (8) Paul Bunyan;

      (9) Associated tools and Minor equipment;

      (10) Ship Staff/Support manning assistance;

      (11) Rigid Hull Inflatable Boat (RHIB); and,

      (12) Supervision.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) Weather conditions not to exceed Sea State 2; and,

      (2) Day light operations.

3. Standard. Conduct Submarine Towed Array Operations At Sea, to include:

a. Deploying Towed Array to include:

(1) Transferring Stub Cable from Submarine to TAV;

(2) Removing the dummy load and connecting the Stub cable to the Tow Cable;

(3) Deploying Stub cable with Drogue Floats; and,

(4) Deploying the Tow cable and Array.

b. Recovering the Array to include:

(1) Grapnel recovery of Drogue Floats;

(2) Grapnel recovery of Tow Cable;

(3) Seine gate Recovery;

(4) Disconnecting Tow From Stub Cable and reconnecting the dummy load;

(5) Transferring the Stub cable form The TAV to the Submarine; and,

(6) Winding the Tow and Array cables onto the Drum assembly or Reeler.

c. Carrying out Emergency Operating Procedures, to include: (1) Performing emergency breakaway; and,

(2) Conducting a controlled release.

4. References.

A1, A2, A6, A7, C1 and C2.

5. Specification Task Numbers.

T0003, T0004, T0005 and T0006, T0010, T0011, T0012, T0013, T0014, T0015, T0016, T0017, T0018, T0019, T0020.

6. Supporting Skills.

Nil.

7. Supporting Knowledge.

K0016-K0035.

8. Training Limitations.

a. Availability of an Operational Submarine or suitable replacement vessel;

b. Availability of QHM assets/Crane Services; and,

c. Weather.

PO 003 CONDUCT FOREIGN PORT TOWED ARRAY OPERATIONS

1. Performance: Conduct Foreign Port Towed Array Operations.

2. Conditions:

   a. Given:

      (1) Tulsa Reeler;

      (2) Towed Array reels;

      (3) Towed Array Handling equipment;

      (4) Paul Bunyan;

      (5) Containerized System;

      (6) 53 ft dual axle Flat Bed Trailer;

      (7) Shipping agent;

      (8) DSP; and,

      (9) Stub Cable.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) In Any Conditions.

3. Standard. Conduct Foreign Port Towed Array Operations to include:

   a. Preparing Towed Array equipment for Foreign Port Deployment; and,

   b. Conducting a Port Survey.

4. References.

   A2, B1, D1, D2.

5. Specification Task Numbers.

   T0001, T0002, T0006, T0007, T0009.

6. Supporting Skills.

   Nil.

7. Supporting Knowledge.

   K0001 – K0011.

8. Training Limitations.

   a. Availability of an operational submarine or suitable replacement vessel;

   b. Availability of a suitable TAV and Jetty;

   c. Availability of QHM assets; and,

   d. Weather.

CHAPTER 3: ASSESSMENT OF TRAINEE

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY THE COMMANDANT / COMMANDING OFFICER OF CANADIAN FORCES NAVAL OPERATIONS SCHOOL

GENERAL

1. This is a pass/fail course. All Performance Checks (PCs) are based on Performance Objectives (POs) in Chapter 2 of this QSP and must be successfully completed to achieve this qualification.

PERFORMANCE CHECKS

2. Summative evaluation for this course is in the form of PCs conducted at the termination training for each PO. As per MARCORDS 9-60, PCs are tests to assess the accomplishment of the performance under the conditions and to the standards stated in the PO. While the majority of the assessments are practical, other forms of evaluation are conducted where practical assessment is not feasible or possible.

3. Conduct of any PC will be IAW SSO 307 promulgated by the commandant, Canadian Forces Naval Operations School Halifax.

4. Supplemental PCs must be in the same format and have the same standard as the original test. PCs may contain more than one part to facilitate different methods of assessment or to modularize particularly extensive performance requirements. Unless otherwise stipulated, failure of one part of a PC only requires retest on that part and does not require the trainee to be retested on the entire PC.

RESPONSIBILITY FOR TESTING

5. Standards cell provides a measure of quality control by confirming that POs are achieved according to the standards delineated in the QSP. They are responsible for the design, administration and scoring of all Performance Checks. Enabling Checks will be the responsibility of the Instructional Staff.

PROGRESS MONITORING

6. Student Progress monitoring including unsatisfactory progress IAW CFNOS SO 307

TRAINING REVIEW BOARD

7. Training Review Board requirements are contained in MARCORD 9-60 Volume 1 and IAW CFNOS SO 307.

COURSE REPORTS

8. Course reports (CF 377) will be prepared IAW DAOD 5031-9 and NTI 03/2007. The narrative in block 17 will indicate the trainee's strengths and weaknesses in relation to specific POs and provide information of use to the trainee's Commanding Officer and

Career Manager. Training limitations or incomplete training must be identified on the course report.

CHAPTER 4: LESSON SPECIFICATIONS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY THE COMMANDANT / COMMANDING OFFICER OF CANADIAN FORCES NAVAL OPERATIONS SCHOOL

EO 001.01 IDENTIFY TOWED ARRAY EQUIPMENT

1. Performance: Identify Towed Array Equipment.

2. Conditions:

   a. Given:

      (1) Class room and teaching aids; and,
      (2) References.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) N/A.

3. Standard. Identify Towed Array Equipment, to include:

   a. “Dry end” components;

   b. “Wet end” components; and,

   c. Handling Equipment.

4. Time.

   5 Periods

5. Method / Media.

   a. Lecture/Demo - 4 Periods

   b. EC - 1 Period

6. References.

   A2, A3, A4, A5, C1, C2 and D1.

7. Training Aids.

   a. Power Point presentation;

   b. Diagrams.

8. Test Details.

This EO will be tested IAW EC 001.1.

9. Remarks.

Nil.

10. Teaching Points.

	Description	Method	Time	Reference
TP1	Dry End components • Purpose • Capabilities and Limitations • Location of ▪ TATU ▪ Hydraulic Release ▪ Consoles ▪ Processor Drawers	Lecture	1	B2

	Description	Method	Time	Reference
TP2	Wet End components • Purpose • Capabilities and Limitations • Location and Dimensions of: ▪ Umbilical Cable ▪ Tow box ▪ Stub Cable ▪ Tow Cable ▪ Array ▪ Rope drogue ▪ Dummy Load Cone ▪ Line transfer Kit ▪ Float Drogue	Lecture	1	A1, A2, A3
TP3	Handling Equipment • Purpose • Controls and Indicators • Basic Operation of equipment	Lecture/Demo	2	C1 and C2

ENABLING CHECK – EO 001.01

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom

Arrangements and Procedures

2. EC 001.01 will be a written test based on content from EO 001.01.

3. Pass mark will be 70% 4.

   Time required: 1 x 50 min

   period

5. References will not be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.01 will be a written test based on content from EO 001.01.

2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s).

6. Read each question carefully, and all responses fully, before answering the question.

7. If you have any questions during the exam, raise your hand.

8. References will not be provided.

9. Pass mark is 70%

10. You will have 1 x 50 min period to complete this exam.

11. Do you have any questions before you start?

EO 001.02

1. Performance: Operate Towed Array Handling Equipment

2. Conditions:

   a. Given:

      (1) Tulsa Reeler;
      (2) TA Reels;
      (3) Containerized Winch System;
      (4) DSP;
      (5) Crane, Operator and Rigger;
      (6) Generator;
      (7) Closing Tool;
      (8) Opening tool;
      (9) References; and,
      (10) Supervision.

   b. Denied:

      Nil.

   c. Environmental:

      (1) All weather conditions alongside; and,

      (2) IAW Crane restrictions.

3. Standard. Operate Towed Array Handling Equipment, to include:

   a. Tulsa Reeler;

   b. Containerized Winch System;

   c. DSP;

   d. Generator;

   e. Closing tool;

   f. Opening tool;

   g. Loading out the Towed Array Vessel; and,

   h. Unloading the Towed Array Vessel.

4. Time.

21 Periods

5. Method / Media.

   a. Lecture/Demo 20 Periods; and,

   b. EC 1 Period.

6. References.

   A5, C1 and C2.

7. Training Aids.

   Operational Tools and Equipment.

8. Test Details.

   This EO will be tested IAW EC 001.2.

9. Remarks.

   Nil

10. Teaching Points.

	Description	Method	Time	Reference
TP1	Operation of the Tulsa Reeler: • Functional Description • Pre Start Routine • Reel Loading/Unloading • Flash-up • Reel operation • Monitoring Controls and Indicators; and • Securing	Lecture/ Demo	3	C1

	Description	Method	Time	Reference
TP2	Operation of the Containerized Winch System • Functional Description • Pre Start Routine • Reel Loading/Unloading • Flash-up • Reel operations • Monitoring controls and Indicators; and • Securing	Lecture/ Demo	3	C2
TP3	Transfer Array and Tow Cable from One Reel to Another	Lecture/ Demo	3	C2
TP4	Operation of the Generator • Functional Description • Pre start routine • Flash-up • Operation; and • Securing	Lecture/ Demo	2	C2
TP5	Operation of the opening tool, to include: • Inspection of tool • Operation • Equipment damage • Proper securing of tool Care and maintenance	Lecture/ Demo	1	A2

	Description	Method	Time	Reference
TP6	Operation of the closing tool • Inspection of tool • Operation • Equipment damage • Proper securing of tool Care and maintenance	Lecture/ Demo	1	A2
TP7	Load out the Towed Array Vessel • Inspection ▪ Working Deck • Size • Risers • Hard points • Freeboard • Adjacent Compartments • Loading plan • Support requirements ▪ Cranes ▪ Riggers ▪ Low bed • Securing arrangements • Safety Hazards • Precautions	Lecture/ Demo	5	A2

	Description	Method	Time	Reference
TP8	Unload the Towed Array Vessel Support requirements ▪ Cranes ▪ Riggers ▪ Low bed	Lecture/ Demo	1	A2

ENABLING CHECK – EO 001.02

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom

Arrangements and Procedures

2. EC 001.02 will be a Written test based on content from EO 001.02.

3. Pass mark will be 70% 4.

   Time required: 1 x 50 min

   period

5. References will not be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.02 will be a Written test based on content from EO 001.02.

2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s).

6. Read each question carefully, and all responses fully, before answering the question.

7. If you have any questions during the exam, raise your hand.

8. References will not be provided.

9. Pass mark is 70%

10. You will have 1 x 50 min period to complete this exam.

11. Do you have any questions before you start?

EO 001.03 PERFORM STUB CLIP ON/OFF ROUTINES

1. Performance: Perform Stub Clip on / off Routines.

2. Conditions:

   a. Given:

      (1) TA Reels;
      (2) DSP;
      (3) Crane, forklift and Riggers;
      (4) STAHT Equipment;
      (5) Stub and Tow cables and Array;
      (6) Diver Support;
      (7) References; and,
      (8) Supervision.

   b. Denied:

      Nil.

   c. Environmental:

      (1) All weather conditions alongside; and,

      (2) IAW local crane restrictions.

3. Standard. Perform Stub Clip on / off Routines, to include:

   a. Participating in fitting the stub cable;

   b. Participating in removing the stub cable;

   c. Operating the DSP; and,

   d. Securing the DSP.

4. Time.

   a. 17 Periods

5. Method / Media.

   a. Lecture/Demo/Practical – 9 Periods;

   b. EC - 1 Period; and,

   c. PC – 7 Periods.

6. References.

   A2, A6, A8, A10 D1.

7. Training Aids.

   Equipment and tools.

8. Test Details.

   This EO will be tested IAW EC 001.3.

9. Remarks.

10. NilTeaching Points.

	Description	Method	Time	Reference
TP1	Preparation of the DSP	Lecture	1	A8
TP2	Trimming Down the Submarine	Lecture	1	A10
TP3	Placement of the DSP	Lecture	1	A2
TP4	Raising the Upper Tow Box	Lecture	1	A2
TP5	Testing of the Hydraulic Release Rams	Lecture	.5	A2
TP6	Connection of the Stub cable	Lecture	.5	A2
TP7	Loop Resistance Checks	Lecture	1	A2
TP8	Confidence Checks	Lecture	.5	A2
TP9	Removal of the Stub Cable	Lecture	1	A2
TP10	Securing of DSP	Lecture	1	A8

ENABLING CHECK – EO 001.03

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

6. Classroom

Arrangements and Procedures

7. EC 001.02 will be a written test based on content from EO 001.02.

8. Pass mark will be 70% 9.

   Time required: 1 x 50 min

   period

10. References will not be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

12. EC 001.02 will be a written test based on content from EO 001.02.

13. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.

14. Do not open the booklet until you are told to do so.

15. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

16. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s).

17. Read each question carefully, and all responses fully, before answering the question.

18. If you have any questions during the exam, raise your hand.

19. References will not be provided.

20. Pass mark is 70%

21. You will have 1 x 50 min period to complete this exam.

22. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. The instructor shall inform the trainee of the assessment criteria.

PART IV – ENABLING CHECK LIST

CHECK LIST	YES	NO	COMMENTS
Trimming Down			
DSP Preparation			
DSP Placement			
Upper Tow Box removal			
Hydraulics Checks			
Stub / Umbilical Connection			
Electrical checks			
Securing Stub			
DSP securing			

PERFORMANCE CHECK – PO 001

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom
2. Logistic Support
3. TA Handling equipment; and
4. References

Arrangements and Procedures

5. PC 001 will be a Practical test based on content from PO 001.
6. Pass mark will be 70% 7.

   Time required: 5 x 50 min

   period

8. References will not be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. PC 001 will be a Practical test based on content from PO 001.
2. The Trainee shall operate the following equipment under test conditions:
   a. Tulsa Reeler (MARENG and ETECH)
   b. Containerized Winch System (MARENG and ETECH)
   c. Portable Generator (All)
   d. Opening and Closing Tools (NE and NW TECHS)
3. Work carefully following all safety requirements.
4. If you have any questions during the test, raise your hand.
5. References will not be provided.
6. Pass mark is 70%
7. You will have 5 x 50 min period to complete this test.
8. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. The instructor shall inform the trainee of the assessment criteria.

PART IV – PERFORMANCE CHECK LIST

CHECK LIST	YES	NO	COMMENTS
Tulsa Reeler Operational Checks			
Pre Operational Checks			
Ensure the reeler is properly secured to the operating platform			
Ensure the unit is grounded properly			
Ensure the Exhaust Muffler Assembly is bolted back on to the engine block (Normally removed to ship or sling the reeler)			
Ensure cables to the Deadman's switch and hand control unit are in good operating condition			
Engine Lubrication – check crankcase oil level ("F" mark on the dip stick)			
Engine Coolant – ensure engine is filled with 50/50 mixture of ethylene glycol base antifreeze and soft water (fill coolant recovery bottle ½ full)			
Fan Belt Tension – check engine fan belt tension (deflection of 3/8" 5/8")			
Fuel Supply – Check that an ample supply of diesel fuel is available to the engine fuel system			
Battery and cables – inspect battery and cables, posts should be clean and cables attached properly. Ensure battery is fully charged and electrolyte level is correct. Add distilled water as required.			
Hydraulic System – ensure hydraulic tank is ¾ full. Inspect hoses and check for leaks, ensure no water in the tank or system			

CHECK LIST	YES	NO	COMMENTS
(sight glass)			
Ensure no obstructions are blocking air flow to the engine cooling fan			
All moving parts should be free of obstructions (i.e. Rags, tools, etc…)			
Ensure “ABC” class fire extinguisher is available.			
Ensure STAHT tool box is available (tool box to be mustered regularly)			
Start up and Operating/Monitoring			
“Auto/Manual/Off” switch should be placed in the manual position			
Push and hold the “start/stop” switch. When the engine starts, release the switch **or after 5 sec** of cranking (Check for Leaks)			
After the generator is running, adjust the voltage to 480 VAC output with the “Voltage adjust Knob”			
Make the “Main Power Breaker” inside enclosure			
Turn the “Tension and Speed” pots fully counter clockwise. On the pendant, turn the “Load /Run” switch to the Run position			
Make the service connect/disconnect switch on the main enclosure and control panel			
Ensure the “PWR/TENS” switch is in the “PWD” position. Push the Hydraulic start button on the control panel ensuring proper phase is selected. Air should be blowing through the heat exchanger.			

CHECK LIST	YES	NO	COMMENTS
(Check for Leaks)			
Ensure the Manual Brake is not engaged			
The Reeler is now ready for Operation			
During the operation constant vigilance is required in monitoring the unit for: a. over crane advisory lamp b. high coolant temp lamp c. overspeed advisory lamp d. low oil pressure advisory lamp e. RPM sensor loss lamp f. A/C voltmeter g. A/C ammeter h. A/C frequency meter i. Oil pressure gauge j. D/C ammeter k. Hour Meter l. Engine coolant temp gauge; and m. Secondary fuel tank detect (light on indicates internal fuel tank leak)			
Securing Checks			
Push the “stop” button on the control panel. Ensure the “PWD/TENS” switch is in the “TENS” position			
Break the Service			

CHECK LIST	YES	NO	COMMENTS
connect/disconnect switch			
Put the “Start/ Stop” switch to the stop position until the engine comes to a complete stop			
Secure the grounding cable			
If the Reeler is left outside for an extended period, cover the unit with it’s protective cover			
115Volt 60 Cycle Generator Checks			
Pre operational checks			
Ensure Grounding strop is connected			
Ensure Lube oil level is correct			
Ensure Fuel tank is full			
Do not start in enclosed area. Ensure adequate air supply			
Ensure generator is free from obstructions			
Do not operate under wet conditions without a protective tarp			
Start up and Operation/Monitoring			
Ensure “On/Off” switch on Electrical outlet box is in the “Off” position			
Ensure Fuel supply is turned on			
Put “Start/Stop” switch to start position and start engine			
Check for Leaks			
Turn “On/Off” switch on the electrical outlet box to the “On” position			

CHECK LIST	YES	NO	COMMENTS
Do not overload the unit			
Never add fuel to the unit while it is running			
Securing Checks			
Turn the “On/Off” switch on the electrical outlet box to the “Off” position			
Put the fuel throttle valve to the “No Fuel “ position			
Turn the “Start/Stop” switch to the “Stop” position			
Umbilical Cable Closing / Opening Tool Operational Checks			
Pre operational checks - Check all joints are lubricated and move freely - Check all circlips / joints are intact			
Operation - Open the closing tool - Place the closing tool over the umbilical cable and the stub cable connectors with the handles perpendicular to the cables with the tapered end on the stub cable - Shut the claps on the tool (2) - Ensure cables are lined up - Draw the handles together to make the connection			

CHECK LIST	YES	NO	COMMENTS
- Attach the two clam shell locks			
Securing checks Ensure the closing tool is well lubricated and operational			

ASSESSORS / TACTICS REMARKS:

OVERALL ASSESSMENT:

<table>
<tr><td colspan="2">Are the performances passed?</td></tr>
<tr><td>YES</td><td>NO</td></tr>
<tr><td>Are __ other performance factors satisfactory or above?</td><td rowspan="3">FAIL</td></tr>
<tr><td>YES</td></tr>
<tr><td>PASS</td></tr>
</table>

_____ _____

STUDENTS SIGNATURE STAFF ASSESSOR

THIS FORM TO BE RETAINED ON COURSE FILES IAW CFNOSSO'S

EO 002.01 DEPLOY A TOWED ARRAY

1. Performance: Deploy a Towed Array.

2. Conditions:

   a. Given:

      (1) Classroom;
      (2) TAV;
      (3) Submarine or suitable substitute vessel with Stub cable fitted;
      (4) RHIB;
      (5) Suitable Deployment / Recovery site; (6) Supervision to include full STAHT; and,
      (7) References.

   b. Denied:

      Nil

   c. Environmental:

      N/A

3. Standard. Deploy a Towed Array, to include:

   a. Transferring a Stub Cable;

   b. Removing the Dummy Load;

   c. Deploying the Stub cable with Floats; and,

   d. Deploying the Tow cable and Array.

4. Time.

   9 Periods

5. Method / Media.

   a. Lecture/Demo/Practical – 8 Periods; and,

   b. EC – 1 Period.

6. References.

   A2, A11.

7. Training Aids.

a. STAHT Equipment, including Stub Cable and Dummy Load Transfer Kit;

b. TAV; and,

c. Submarine or Substitute.

8. Test Details.

This EO will be tested IAW EC 002.1.

9. Remarks.

Nil.

10. Teaching Points.

	Description	Method	Time	Reference
TP1	Responsibilities of: • Submarine CO ▪ When Connected ▪ Not Connected • STAHO ▪ When Connected ▪ Not Connected • Casing Party • TAV Master • TASR • TA NET • Mar Eng • E Tech • Line Handlers • RHIB crew	Lecture	1	A2

TP2	Deployment Methods to include; • Primary (one step / line transfer); and • Secondary method (two step / drogue transfer	Lecture	1	A2
TP3	Stub Cable transfer from Submarine to TAV	Lecture/ Demo/ Practical	1	A2
TP4	Dummy load removal and connection of the Stub cable to the Tow Cable	Lecture/ Demo/ Practical	2	A2
TP5	Stub cable with Drogue Floats deployment; and	Lecture/ Demo/ Practical	2	A2
TP6	Tow cable and Array deployment	Lecture/ Demo/ Practical	1	A2

ENABLING CHECK – EO 002.01

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom

Arrangements and Procedures

2. EC 001.01 will be a written test based on content from EO 002.01.

3. Pass mark will be 70% 4.

   Time required: 1 x 50 min

   period

5. References will not be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 002.01 will be a written test based on content from EO 002.01.

2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s).

6. Read each question carefully, and all responses fully, before answering the question.

7. If you have any questions during the exam, raise your hand.

8. References will/will not be provided.

9. Pass mark is 70%

10. You will have 1 x 50 min period to complete this exam.

11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Nil

PART IV – ENABLING CHECK LIST

CHECK LIST	YES	NO	COMMENTS
Responsibilities			
Deployment Methods			
Dummy Load Removal			
Float Drogue Deploy			

EO 002.02 RECOVER A TOWED ARRAY

1. Performance: Recover a Towed Array.

2. Conditions:

   a. Given:

      (1) Classroom;
      (2) TAV;
      (3) Submarine or suitable substitute vessel with Stub cable fitted;
      (4) RHIB;
      (5) Suitable Deployment / Recovery site; (6) Supervision to include full STAHT; and,
      (7) References.

   b. Denied:

      Nil.

   c. Environmental:

      All weather conditions not exceeding Sea State 2.

3. Standard. Recover a Towed Array, to Include:

   a. Recovering the Stub cable with Floats;

   b. Recovering the Tow cable and Array;

   c. Installing the Dummy Load;

   d. Transferring a Stub Cable; and,

   e. Reversing the TA.

4. Time.

   a. 10 Periods

5. Method / Media.

   a. Lecture/Demo/Practical – 9 Periods; and,

   b. EC – 1 Period.

6. References.

A2.

7. Training Aids.

   a. STAHT Equipment including Stub Cable and Dummy Load Line Transfer Kit;

   b. TAV; and,

   c. Submarine or Substitute.

8. Test Details.

   This EO will be tested IAW EC 002.2.

9. Remarks.

   Nil.

10. Teaching Points.

	Description	Method	Time	Reference
TP1	Responsibilities of: • Submarine CO ▪ When Connected ▪ Not Connected • STAHO ▪ When Connected ▪ Not Connected • Casing Party • TAV Master • TASR • TA NET • Mar Eng • E Tech • Line Handlers	Lecture	1	A2
	▪ RHIB crew			

TP2	Recovery Methods to include; • Primary (one step / line transfer); and ▪ Seine Gate • Secondary method (two step / drogue transfer • Grapnel	Lecture	1	A2
TP3	Stub cable with Floats recovery	Lecture/ Demo/ Practical	1	A2
TP4	Tow cable and Array recovery	Lecture/ Demo/ Practical	1	A2
TP5	Dummy Load Installation	Lecture/ Demo/ Practical	2	A2
TP6	Stub Cable Transfer	Lecture/ Demo/ Practical	2	A2
TP7	Array Reversal	Lecture	1	A2

ENABLING CHECK – EO 002.02

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom

Arrangements and Procedures

2. EC 002.02 will be a written test based on content from EO 002.02.

3. Pass mark will be 70% 4.

   Time required: 1 x 50 min

   period

5. References will not be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 002.02 will be a written test based on content from EO 002.02.

2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s).

6. Read each question carefully, and all responses fully, before answering the question.

7. If you have any questions during the exam, raise your hand.

8. References will/will not be provided.

9. Pass mark is 70%

10. You will have 1 x 50 min period to complete this exam.

11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

PART IV – ENABLING CHECK LIST

CHECK LIST	YES	NO	COMMENTS
Responsibilities			
Float Drogue Recovery			
Dummy Load Installation			

EO 002.03 PERFORM TOWED ARRAY EOPs

1. Performance: Perform Towed Array EOPs.

2. Conditions:

   a. Given:

      (1) TAV;
      (2) Submarine or suitable substitute vessel with Stub cable fitted;
      (3) RHIB;
      (4) Suitable Deployment / Recovery site; (5) Supervision to include full STAHT; and,
      (6) References.

   b. Denied:

      Nil.

   c. Environmental:

      Weather conditions deemed acceptable by the Submarine CO and STAHO.

3. Standard. Perform Towed Array EOP's, to include:

   a. Conducting Emergency Break away;

   b. Conducting Man over board; and,

   c. Conducting Controlled Release.

4. Time.

   a. 10 Periods

5. Method / Media.

   a. Lecture/Demo/Practical – 2 Periods;

   b. EC – 1 Period; and,

   c. PC – 7 Periods.

6. References.

   A2, A10, A11.

7. Training Aids.

a. STAHT Equipment;

b. TAV; and,

c. Submarine or Substitute.

8. Test Details.

This EO Will be Tested IAW PC 002.

9. Remarks.

Nil.

10. Teaching Points.

	Description	Method	Time	Reference
TP1	TA EOP's • Emergency Break away procedures ▪ Connection on Deck ▪ Connection in the water • Man over board ▪ Method of recovery	Lecture/ Demo/ Practical	1	A2, A11
TP2	Controlled Release • Environmental considerations • Equipment safety • Personnel Safety	Lecture	1	A2, A10

ENABLING CHECK – EO 002.03

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom

Arrangements and Procedures

2. EC 002.03 will be a written test based on content from EO 002.03.

3. Pass mark will be 70% 4.

   Time required: 1 x 50 min

   period

5. References will not be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 002.03 will be a written test based on content from EO 002.03.

2. If you have any questions during the exam, raise your hand.

3. References will not be provided.

4. Pass mark is 70%

5. You will have 1 x 50 min period to complete this exam.

6. Do you have any questions before you start?

PERFORMANCE CHECK – PO 002

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Functional Towed Array Equipment
2. Barge and Tugs
3. Submarine or alternate

Arrangements and Procedures

4. EC 002.03 will be a Practical test based on content from EO 002.03.
5. Pass mark will be 70%
6. Time required: 7 x 50 min period
7. References will not be required other than procedure cards for the STAHO and TASR candidates.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

8. EC 002.03 will be a Practical test based on content from EO 002.03.
9. If you have any questions during the exam, raise your hand.
10. References will not be required other than procedure cards for the sTAHO and TASR candidates.
11. Pass mark is 70%
12. You will have 7 x 50 min period to complete this exam.
13. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

Nil

PART IV – PERFORMANCE CHECK LIST

CHECK LIST	YES	NO	COMMENTS
Performance Check STAHO			**Deployment**
Ensure correct safety brief is given (to include individual / collective safety and safety responsibilities of various members of the team)			

Communications with TASR, Submarine and RHIB. Ensure comm.. checks, correct frequencies, call signs and communications security			
Interaction / Liaison with Halifax traffic / Harbour traffic			
Ensure Stub is powered down			
Correct positioning of vessels initially to prepare for passing of the stub cable			
Oversee passing of the stub cable (including vessel positioning) to ensure the job of the line handlers is facilitated			
Oversee connection of the Stub cable to the Tow cable (keep the vessels correctly positioned)			
Positioning of the vessels to commence deployment of the Array (ensuring appropriate tension and arc is maintained through out the deployment)			
Complete all steps as outlined in STAHM Ref A2			
Overall control and supervision of safety though out			
Student must achieve 80% pass mark			
Student Name		**S/N**	

ASSESSORS / TACTICS REMARKS:

OVERALL ASSESSMENT:

<table>
<tr><td colspan="2">Are the performances passed?</td></tr>
<tr><td>YES</td><td>NO</td></tr>
<tr><td>Are __ other performance factors satisfactory or above?</td><td rowspan="3">FAIL</td></tr>
<tr><td>YES</td></tr>
<tr><td>PASS</td></tr>
</table>

_____ _____

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CHECK LIST	YES	NO	COMMENTS
Performance Check TASR			**Deployment**
Conduct Barge Safety Brief prior to arrival at the Deployment area, to include: - The Task - Who is in charge ○ TASR ○ OIC ○ TAV Master ○ Sub CO - Responsibility for Safety ○ TASR ○ OIC ○ TAV Master ○ Sub CO - Individual task assignments - Deck safety - Hazards - Risk management - Sign the safety brief sheet			
Oversee the availability and placement of barge resources for deployment			
Ensure the rope drogue securing tie is cut (Tulsa Reeler)			
Know calls signs and selected channels and monitor communications with STAHO as required			
Co-ordinate stub transfer			

CHECK LIST	YES	NO	COMMENTS
Direct cutting of the stub cable tie downs from the Sub using proper hand signals			
Over see the maintaining of correct tension on the stub cable			
Over see the electrical connection of the Stub and Tow cables (ensuring TATU is powered down)			
Over see the deployment of the Tow and Array (ensuring correct bend radius)			
Student must achieve 70% pass mark			
Student Name		**S/N**	

ASSESSORS / TACTICS REMARKS:

OVERALL ASSESSMENT:

<table>
<tr><td colspan="2">Are the performances passed?</td></tr>
<tr><td>YES</td><td>NO</td></tr>
<tr><td>Are __ other performance factors satisfactory or above?</td><td rowspan="3">FAIL</td></tr>
<tr><td>YES</td></tr>
<tr><td>PASS</td></tr>
</table>

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CHECK LIST	YES	NO	COMMENTS
Performance Check Marine Electrician			**Deployment**
Ensure Reeler is properly grounded			
In Co-operation with the Mar Eng ensure the fluids are topped up			
Ensure electrical connections and lighting are rigged as necessary and working			
Verify the operation of the 115v generator			
Ensure electrical cables are safely laid out			
Ensure there are no obstructions present prior to starting the Tulsa Reeler / Power Pack And Drum unit			
Student must achieve 70% pass mark			
Student Name		**S/N**	

ASSESSORS / TACTICS REMARKS:

OVERALL ASSESSMENT:

<table>
<tr><td colspan="2">Are the performances passed?</td></tr>
<tr><td>YES</td><td>NO</td></tr>
<tr><td>Are __ other performance factors satisfactory or above?</td><td rowspan="3">FAIL</td></tr>
<tr><td>YES</td></tr>
<tr><td>PASS</td></tr>
</table>

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STUDENTS SIGNATURE STAFF ASSESSOR

THIS FORM TO BE RETAINED ON COURSE FILES IAW CFNOSSO'S

EO 003.01 PLAN FOREIGN PORT TOWED ARRAY OPERATIONS

1. Performance: Plan Foreign Port Towed Array Operations.

2. Conditions:

   a. Given:

      (1) Tulsa Reeler;
      (2) TA Reels;
      (3) Containerized TA System;
      (4) Diver Support Platform (DSP);
      (5) Crane, Riggers, Forklift;
      (6) 53ft, Dual Axle, Flatbed trailer;
      (7) 5 tonne box truck;
      (8) Paul Bunyan;
      (9) Stub, Tow and Array cables; (10) References; and,
      (11) Supervision.

   b. Denied:

      Nil.

   c. Environmental:

      (1) All weather conditions alongside; and,

      (2) IAW local crane restrictions.

3. Standard. Plan Foreign Port Towed Array Operations, to include:

   a. Conducting administration;

   b. Administer transport Load out procedures; and,

   c. Documentation requirements.

4. Time.

   5 Periods

5. Method / Media.

   a. Lecture – 4 Periods; and,

   b. EC – 1 Period.

6. References.

   A2, B1, D1.

7. Training Aids.

   Equipment and tools.

8. Test Details.

   This EO will be tested IAW EC 003.1.

9. Remarks.

   Nil.

10. Teaching Points.

	Description	Method	Time	Reference
TP1	Administration • Timings • Vehicle requirements • Transport Vessel requirements	Lecture	1	B1, D1
TP2	Loading out Transport • Tulsa Reeler • Reels • Containers ▪ 10 ft ▪ 20 ft • DSP • Paul Bunyans • Stub Stowage Box; • Diver Support Equipment and • Spare reel supports	Lecture	1	A9, D1

	Description	Method	Time	Reference

TP3	Documentation requirements
	• Hazmat (usually sent on sub)
	  ▪ MSDS
	• Packing
	  ▪ Bill of lading
	• Customs
		Lecture	2	B1

ENABLING CHECK – EO 003.01

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom

Arrangements and Procedures

2. EC 003.01 will be a written test based on content from EO 003.01.

3. Pass mark will be 70%

4. Time required: 2 x 50 min periods

5. References will/will not be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 003.01 will be a written test based on content from EO 003.01.

2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s).

6. Read each question carefully, and all responses fully, before answering the question.

7. If you have any questions during the exam, raise your hand.

8. References will/will not be provided.

9. Pass mark is 70%

10. You will have 2 x 50 min periods to complete this exam.

11. Do you have any questions before you start?

EO 003.02 CONDUCT PORT SURVEY

1. Performance: Conduct Port Survey.

2. Conditions:

   a. Given:

      (1) Internet;
      (2) Check off List of requirements (available from MOG5 TASR);
      (3) Representative Port, complete with shore service facilities and one or more prospective TAV's;
      (4) Camera; and, (5) Tape Measure.

   b. Denied:

      Nil

   c. Environmental:

      All Weather Conditions.

3. Standard. Conduct a Port Survey, to include:

   a. Assessing potential tug service providers;

   b. Assessing Shore Site; and,

   c. Assessing the Operation Site.

4. Time.

   21 Periods

5. Method / Media.

   a. Lecture/Demonstration/Practical – 14 Periods; and,

   b. PC – 7 Periods

6. References.

   D2.

7. Training Aids.

   a. Internet;

   b. Tape Measure; and,

   c. Camera.

8. Test Details.

This EO will be tested during PC 003.

9. Remarks.

a. Note: The Towed Array Survey Team does not have the authority to contract the vessel. The Chandler or Supply Purchaser only, sets up the contract.

10. Teaching Points.

	Description	Method	Time	Reference
TP1	Internet search of at least three potential Tug companies	Lecture	1	
TP2	Contact Potential companies	Lecture	1	
TP3	Visit each company and Take Pictures of the vessels	Lecture	1	
TP4	Measure the following • Size of working deck ▪ Draft ▪ Freeboard • Bulwarks • Location of any Deck Risers	Lecture/ Demo/ Practical	4	D2
TP5	Identify the following requirements: • Location of company vessel • Master of the Vessel • Propulsion • Ability to weld on the	Lecture	1	D2

	Description	Method	Time	Reference
	deck • Identify Bracket Requirements • Availability of support vessel for seine Gate Ops • Rates			
TP6	Locate and Identify accommodations	Lecture	.5	D2
TP7	Locate Support Assets to include: • Cranes • Forklifts • Staging areas • Lay down areas • Parking areas • Welders • Support vessel (RHIB)	Lecture	.5	D2
TP8	Local Diving Regulations identification	Lecture	1	D2
TP9	Pre departure administration requirements • Arranging Visit Clearances • Arranging TD requirements ▪ International ▪ Domestic	Lecture	2	
TP10	Assessment of Suitable Operational Site	Lecture	2	

	Description	Method	Time	Reference
	• Charted depths/seine gate length of stay calculation • Submerged obstructions • Bottom type • Tides and Currents • Prevailing winds • Traffic • Prevailing/seasonal swells • Restricted areas • Speed considerations for TAV • Speed/time/distance calculations			

ENABLING CHECK – EO 003.02

The requirements of the EC are addressed in PC 003

PERFORMANCE CHECK – PO 003

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Internet
2. Classroom
3. Jetty
4. Potential TAV

Arrangements and Procedures

5. PC 003 will be a Practical assessment based on content from PO 003.
6. Pass mark will be 70% 7.

   Time required: 7 x 50 min

   period

8. References will be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. PC 003 will be a Practical assessment based on content from PO 003.
2. Pass mark is 70%
3. You will have 7 x 50 min period to complete this assessment.
4. Do you have any questions before you start?

PART III – Performance CHECK NOTES PART IV – Performance CHECK LIST

CHECK LIST	YES	NO	COMMENTS
Locate three companies			
Locate accommodations			
Gather information on vessels			
Select the most appropriate vessel			
Locate and Identify Support Agencies			

ASSESSORS / TACTICS REMARKS:

OVERALL ASSESSMENT:

<table>
<tr><td colspan="2">Are the performances passed?</td></tr>
<tr><td>YES</td><td>NO</td></tr>
<tr><td>Are __ other performance factors satisfactory or above?</td><td rowspan="3">FAIL</td></tr>
<tr><td>YES</td></tr>
<tr><td>PASS</td></tr>
</table>

_____ _____

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THIS FORM TO BE RETAINED ON COURSE FILES IAW CFNOSSO'S

ANNEX A – MAIN REFERENCES

A1 = CF or other official Canadian government publication

A2 = Second and subsequent references in any category

B = Official Allied Force or Allied government publication

C = Commercial publication

D = Other publications

A1	C69-827-000/MS-001 (2046 Hand book and Parts list)
A2	C69-827-000/MS-003 (Towed Array handling Manual)
A3	C69-827-000/NG-003 (TA Functional Checks)
A4	C69-827-000/NK-001 (Pre-Deployment Checks)
A5	C69-827-000/NK-002 (Post-Deployment Checks)
A6	C69-827-000/NK-003 (Confidence Checks)
A7	VCSSO's Vol 3
A8	C69-849-000/NY-001 (DSP)
A9	C69-850-000/VS-001 (Stowage and Shipping)
A10	VCSOP's
A11	VCEOP's
B1	SubTass Logistics Directive
B2	BR 8754 Submarine Towed Array Operations
C1	Tulsa Reeler Operating Instructions
C2	Quinetiq Containerized Winch System Manual
D1	Submarine Towed Array Handling Team Check Off Lists
D2	Port Survey Check Off List

ANNEX B – TASK LIST

JBOS/OSS TASK NO.	TASK ELEMENT	TRG / NO TRG	PO
T001	Prepare for Towed Array (TA) operations	Trg	002
T002	Perform logistic support and Plan for domestic and foreign operations	Trg	003
T003	Perform a TA deployment	Trg	002
T004	Perform a TA recovery	Trg	002
T005	Conduct Emergency Operating Procedures	Trg	002
T006	Transfer an array from one reel to another	Trg	001
T007	Preparing Personnel Support Platform	Trg	001
T008	Operating the Tulsa Reeler	Trg	001
T009	Using appropriate, available assets	Trg	001
T010	Packing TA and associated handling equipment	Trg	003
T011	Choosing acceptable primary and secon deployment/recovery sites	Trg	003
T012	Transferring stub cable connector submarinefrom to working platform	Trg	002
T013	Connecting stub cable to tow cable	Trg	002
T014	Paying out tow cable and array into water	Trg	002
T015	Performing seine gate recoveries	Trg	002
T016	Performing grapnel recoveries	Trg	002
T017	Disconnecting stub and tow cable	Trg	002
T018	Transferring stub cable to submarine	Trg	002

T019	Winding in tow cable and array	Trg	002
T020	Reversing the array at sea	Trg	002
T021	Performing Emergency Break-away	Trg	002
T022	Conducting a controlled release	Trg	002

ANNEX C – QUALIFICATION STANDARD AND PLAN WRITING BOARD RECORD OF PROCEEDINGS

Dates: 19-23 Nov 07

Location: Halifax NS

Board Members:

SURNAME	INITIALS	RANK	MOSID	UNIT
Coffen	KH	LCdr	00344	MOG 5 HQ
McDonald	MC	Lt(N)	00344	CFNOS
Sullivan	SMA	SLt	00207	CFNOS
Chapman	M	PO1	00118	MOG 5 HQ
Mayo	D	PO2	00118	HMCS CHICOUTIMI
Berger	E	PO1	00118	CFNOS

Board Objective:

This QSP update mini-board was convened to review the previous version of the OSS and QSP for qualification code AIDI. This QSP was originally completed in 1998, but was never subsequently adopted by DMTE or CFNOS. Since then, Towed Array qualification training has been conducted in a safe but ad-hoc manner by MOG 5, the sole unit supporting Submarine Towed Arrays in the CF to date. It is expected that in the future, other units of the CF may be called upon to support Submarine Towed Arrays, and it was felt important by the current MOG 5 Submarine Towed Array Handling Team to conclude the task of gaining Navy acceptance of a training package.

Specification Task – No Train Decisions:

1. The tasks selected for training are detailed at Annex B. Tasks not selected for training are listed below with substantiation:

TASK SERIAL	TASK ELEMENT	SUBSTANTIATION
	Nil	

Discussions / Decisions:

The older version AIDI OSS was reviewed with the result that all “skills” were moved into the category of “knowledge.” This was done at the insistence of the TDO, who indicated that there is a dispute over the distinction between ‘knowledge’ and ‘skill’ within the TDO community at the present time, with the view that knowledge leads to skill leads to the successful accomplishment of tasks prevailing.

Some of the OSS requirements WRT the knowledge level required were revised upward in recognition of the fact that this qualification would be held commonly for all members of the team from deckhand to STAHO, and due to the fact that this training is preparation for the conduct of an inherently dangerous task.

The QSP review yielded the result that this course required more practical instruction than theoretical classroom instruction, and the EO’s and PO’s were edited to reflect this. EC’s will be written to provide a benchmark for the students prior to undertaking practice runs of the real tasks. The training for the most part will be conducted onboard the TAV, which in the case of Halifax harbour will be a Glen tug and barge combination. The training will be conducted in sheltered water internal to Halifax harbour, likely in the vicinity of York Redoubt, or in the Bedford basin.

There was discussion on the suitability of the course as a basis for qualification of individual members of the STAHT. The decision is that personnel who pass this course will be granted AIDI, however the actual employment of these individuals is subject to the endorsement of the STAHO and TASR for whichever team will employ them, and that, notwithstanding, the AIDI qualification is a requirement for service on a STAHT.

Lessons Learned:

The preformatted QSP template gave us some trouble in terms of both the initial transcription of the 1998 QSP, and in terms of data corruption resulting in several setbacks to production, both during and after the QSP mini-board.Guidance to Instructional Designers:

This board believed firmly that this course requires approximately 2 weeks to conduct, however analysis of the time requirements from the QSP led to the conclusion that in excess of 3 weeks would be required (17 training days). It is important to ensure that the initial estimates are either validated or disproven, and update the QSP accordingly. The live training portion of the course must be supervised by the MOG5 STAHT until sufficient experience with TA Ops is resident elsewhere.

Follow-up Action:

Nil.

Other Considerations (if applicable)

This course exposes students to risk of injury or death. Safety must be paramount at all times in the delivery of the live training.

ANNEX D – QUALIFICATION STANDARD AND PLAN WRITING BOARD SCALAR

By agreement with the TDO, no scalar preparation was undertaken during the QSP Update board.

ANNEX E – ON-JOB PERFORMANCE RECORD

NOT APPLICABLE NOT APPLICABLE

ANNEX F – ACRONYMS / ABBREVIATIONS

ACRONYM / ABBREVIATION	TERM/DEFINITION
TA	TOWED ARRAY
TAV	TOWED ARRAY VESSEL
SEINE GATE	TOWED ARRAY TRAP LINE FOR RECOVERING THE ARRAY
TULSA REELER	MECHANICAL DEVICE USE TO DEPLOY AND RECOVER THE TOW CABLE AND ARRAY
CONTAINERIZED SYSTEM	REPLACEMENT SYSTEM FOR THE TULSA REELER
DSP	DIVER SUPPORT PLATFORM USED TO ATTACH / REMOVE THE STUB CABLE TO / FROM THE SUBMARINE
RHIB	RIDGED HULL INFLATABLE BOAT
EOP	EMERGENCY OPERATING PROCEEDURE

ANNEX G – SUGGESTED SEQUENCING AND TIMINGS

PO/EO	OBJECTIVE	EO	EC	PC	PO
PO 001	**Perform Pre and Post Deployment Activities Alongside for Submarine Towed Array Operations**			7	43
EO001. 1	Identify Towed Array Equipment	4	1		
EO 001.2	Operate Towed Array Equipment	20	1		
EO 001.3	Perform Stub Clip on/off routines	9	1		
PO 002	**Conduct Towed Array Operations at Sea**			7	29
EO002.1	Deploy a Towed Array	8	1		
EO 002.2	Recover a Towed Array	9	1		
EO 002.3	Perform Towed Array EOPs	2	1		
PO 003	**Prepare for Foreign Port Towed Array Operations**			7	26
EO003.1	Prepare Towed Array for Foreign Deployment	4	1		
EO 003.2	Conduct Foreign Port TAV Survey	14			

ANNEX H – TRAINING RESOURCES

In accordance with Chapter 1 paragraph 3 add any extra tables or paragraphs used for Resources in this annex. (**Not** to be deleted.)

ANNEX I – COSTING INFORMATION

Include information about cost in whichever format is applicable.

This annex is optional (**Not** to be deleted).

ANNEX J – COURSE HISTORY

This annex is to record course history issues such as explanations of amendments, pertinent findings, trends, resource problems and modifications. Focus should be given to both excellent and questionable issues of the course. Capturing important course history information is a useful managerial tool for amendment considerations by future course developers.

Note: This annex shall NOT be deleted. This annex must be annotated with every update / modification to this QSP. This is in addition to making the appropriate entry in the Record of Amendments table. Detail as to the nature and reasons for amendments are to be provided here.

1. Formal Training for this Course has not been conducted since the QSP was first developed in 1998. The original course was designed for OBERON Class submarines and was developed in response to a perceived need arising out of several Towed Array handling incidents through the 1990's. The QSP was never implemented, and the last of Canada's OBERONS was decommissioned in 2000. When HMCS WINDSOR commenced operations off the East Coast in 2004, the requirement for Submarine Towed Array support once again presented itself. Maritime Operations Group 5 reintroduced the necessary support skills by engaging a number of qualified personnel from the OBERON era, by undergoing training in the UK with the Royal Navy Submarine Towed Array Handling Team, and by instituting its own program of in-house training ahead of the regeneration of the QSP for this skill set. The MOG 5 team conducted several deployment and recovery operations with HMCS WINDSOR and HMCS CORNER BROOK during the period 2004-2007.

2. In recognition of the need for a Navy-wide course in order to create a wellqualified cadre of individuals who could be drawn upon for membership in safe and proficient TA teams on either the East or West Coasts of Canada, the 1998 TA QSP was reviewed by appropriately qualified and experienced members of the submarine community with the most current experience in VICTORIA class Towed Array support in Nov 2007, and was forwarded for approval ahead of a first course to be taught by CFNOS in fall 2008.

ANNEX K – EVALUATION MATRIX

Note: This annex is optional and **may be** deleted.

EC/PC	Scope Covered	Purpose	How	When	Resources	Limitations	Impact of Limitations